Ports and wiring

What each terminal is for, and the two mistakes that account for most commissioning calls.

The controller has the same terminals on every model in the family. The numbers are printed beside them; the words below are what they do.

The terminals

Terminal	Name	What it is for
1–8	Universal inputs	A switch, a sensor or a 0–10 V signal; the tool decides which
9–14	Digital outputs	Relays, 24 V AC or DC, one amp each
15–16	Analog outputs	0–10 V, for a damper or a valve
17–18	Power	24 V AC, either polarity; the controller does not care

Two mistakes worth knowing

A universal input set to the wrong kind reads plausibly. A 0–10 V sensor wired into an input the tool believes is a switch will read "closed" for ever, and nothing complains. Set the kind before wiring, and confirm it in the tool.

A shared 24 V supply is not shared ground. Two controllers on one transformer share a path that will carry current the wrong way when one of them is switched off. Either give each controller its own transformer, or bond the grounds deliberately — never by accident.

Prove an input

After wiring, change the input by hand and watch the value in the tool. A reading that moves is the only proof that matters; a reading that sits still is worth investigating before the panel goes back on.